Scottish Graduate School of Social Science
Sgoil Cheumnaichean Saidheans Sòisealta na h-Alba

Text Analysis for Social Scientists

SGSSS Training Course

Dr Diarmuid McDonnell

Braw Data Ltd · Gradel Institute of Charity, University of Oxford

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About the Instructor

Dr Diarmuid McDonnell

- Director, **Braw Data Ltd**
- Visiting Fellow, **Gradel Institute of Charity**, University of Oxford
- Research: computational social science, civil society, philanthropy
- Background in quantitative social science and data science

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Course Outline

Time	Session	Format
10:00–10:30	Welcome & Text as Data	Lecture
10:30–11:00	Practical 1: Preprocessing Text	Colab notebook
11:00–11:15	Break	
11:15–11:30	Exploring Text	Lecture
11:30–12:15	Practical 2: Descriptive Text Analysis	Colab notebook
12:15–13:00	Lunch	
13:00–13:30	Topic Modelling	Lecture
13:30–14:45	Practical 3: Topic Modelling	Colab notebook
14:45–15:00	Break	
15:00–15:15	LLMs for Text Analysis	Lecture
15:15–15:50	Practical 4: LLM-Assisted Text Analysis	Colab notebook
15:50–16:00	Wrap-up & Q&A	

Getting Set Up

What you need

- A **Google account** (for Google Colab)
- A web browser (Chrome recommended)

How to access the notebooks

1. Go to the course **README** (link shared on Teams/email)
2. Click the **Open in Colab** badge for your chosen language
3. Materials available in **Python** and **R**
4. Sign in with your Google account — you're ready to go!

ℹ No software installation required — everything runs in the browser.

What Is Quantitative Text Analysis?

Key idea

Treating **text as data** — applying systematic, computational methods to extract meaning from large collections of text.

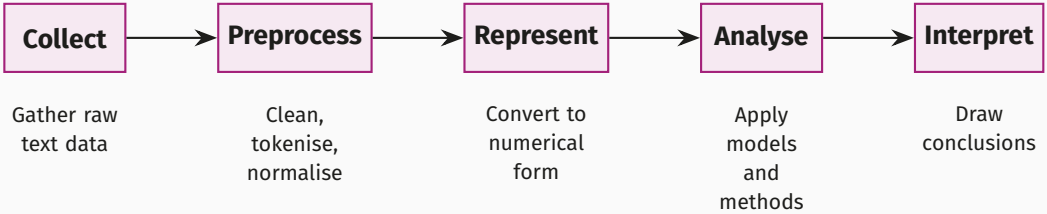
Social science applications:

- Analysing **political speeches** and party manifestos
- Tracking **media coverage** across outlets and over time
- Classifying **policy documents** and legislation
- Studying **online discourse** on social media platforms

Key references

Grimmer & Stewart (2013) · Gentzkow, Kelly & Taddy (2019) · Grimmer, Roberts & Stewart (2022) *Text as Data*

The Text Analysis Pipeline



Today we will work through each of these stages using real-world text data.

Key Concepts

Corpus	A collection of documents (e.g., all speeches by a politician)
Document	A single unit of text (e.g., one speech, one tweet)
Token	An individual occurrence of a word or term
Type	A unique word form in the vocabulary
Vocabulary	The set of all unique types in a corpus

Example

"The cat sat on the mat"

Tokens: The, cat, sat, on, the, mat $\rightarrow$ **6 tokens**

Types: the, cat, sat, on, mat $\rightarrow$ **5 types**

Structured vs Unstructured Text

Unstructured Text

- Raw text documents
- OCR output from scanned pages
- Web scrapes (HTML, forums)
- Free-text survey responses
- Social media posts

Requires significant preprocessing before analysis.

Structured Text

- CSV with a text column
- Labelled corpus (e.g., sentiment)
- Annotated datasets
- Metadata-rich collections
- Pre-processed term–document matrices

Ready (or nearly ready) for quantitative analysis.

Ethics of Computational Text Analysis

- **Consent and privacy** — social media users may not expect their posts to be analysed; consider anonymisation and platform ToS
- **Bias in algorithms** — text models can encode and amplify societal biases present in training data
- **Representativeness of corpora** — whose voices are included and whose are missing? Consider selection effects
- **Transparency and reproducibility** — document preprocessing choices, share code and data where possible

Guiding principle

Computational methods amplify the researcher's reach — ethical scrutiny should be amplified too.

Next Up

Practical 1: Preprocessing Text

- Tokenise and clean real-world text data
- Remove stopwords, punctuation, and noise
- Stem and lemmatise tokens
- Build a document–term matrix

Open the **Practical 1** notebook in Google Colab
(Python or R — your choice!)